

7 (a) (i) It is a region of space in which a force acts on a body. B1

(ii) It is the electric force per unit **positive** charge acting on a small test charge placed at that point. B1
B1

(iii) To eliminate the possibility of magnetic force due to its motion in a magnetic field. B1

b (i) E is the negative of the potential gradient B1
OR

The electric field strength E at a point is numerically equal to the potential gradient at that point and the direction of the E field points in the direction of decreasing potential V .

(ii) 1. Potential gradient at P is negative B1
Electric field $E = -dV/dx$ is positive, hence points in the positive x direction B1

2. From Fig. 7.2 $dV/dx = 0$ at $x = 35$ cm B1
 $E = 0$ B1
Hence $F = eE = 0$ B1

3. -Any charges placed on an isolated conducting sphere **resides** entirely B1
on its outer surface because of repulsive forces between them. Since
there is no charge within the conductor, the **electric field is zero** at every
point **inside** the charged conductor.

$-E = -\frac{dV}{dr} = 0$ implies that all points inside the conductor are at constant B1
electric potential

c (i)	<table><tr><th>x/cm</th><th>V/V</th><th>Vx/Vcm</th></tr><tr><td>13.0</td><td>590</td><td>7670</td></tr><tr><td>21.0</td><td>390</td><td>8190</td></tr></table>	x/cm	V/V	Vx/Vcm	13.0	590	7670	21.0	390	8190	B1 B1
x/cm	V/V	Vx/Vcm									
13.0	590	7670									
21.0	390	8190									

Since Vx not constant, student is incorrect B1

Minimum 2 sets of data needed. x must be read to $\frac{1}{2}$ and V square. Unit of Vx must be given

(ii) The expression would apply only to a single isolated charge.

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However, Fig 7.2 shows the resultant potential between spheres A and B (which is the scalar sum of the potentials due to A and B).

- d (i) By conservation of energy

$$\frac{1}{2}mv_{\min}^2 - eV = 0$$

C1

$$v_{\min} = \sqrt{\frac{2(1.60 \times 10^{-19})295}{9.11 \times 10^{-31}}}$$

$$v_{\min} = 1.02 \times 10^7 \text{ m s}^{-1}$$

A1

- (ii) The electron would approach the spheres along line of minimum electric potential
and cross the A-B line at $x = 35.0 \text{ cm}$

B1

B1